Classification

EEP 596 Computer Vision: Classical and Deep Methods (Autumn 2026)

UNIVERSITY of WASHINGTON

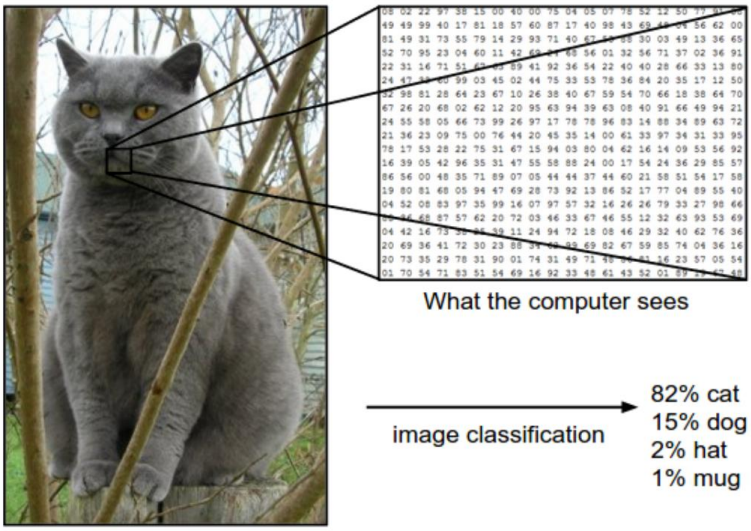
Stan Birchfield

NVIDIA

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试用水印

Image classification



What the computer sees

image classification

82% cat
15% dog
2% hat
1% mug

input
h x w x 3

output
n x 1

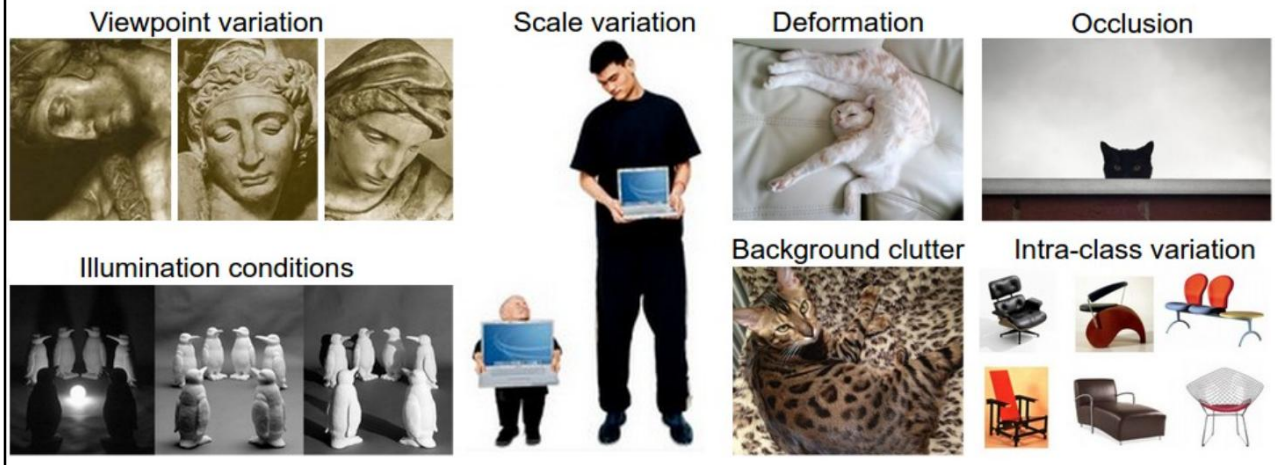
output is PDF
over n classes

PDF = probability
density function
("one hot vector")

https://cs231n.github.io/classification/

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Challenges

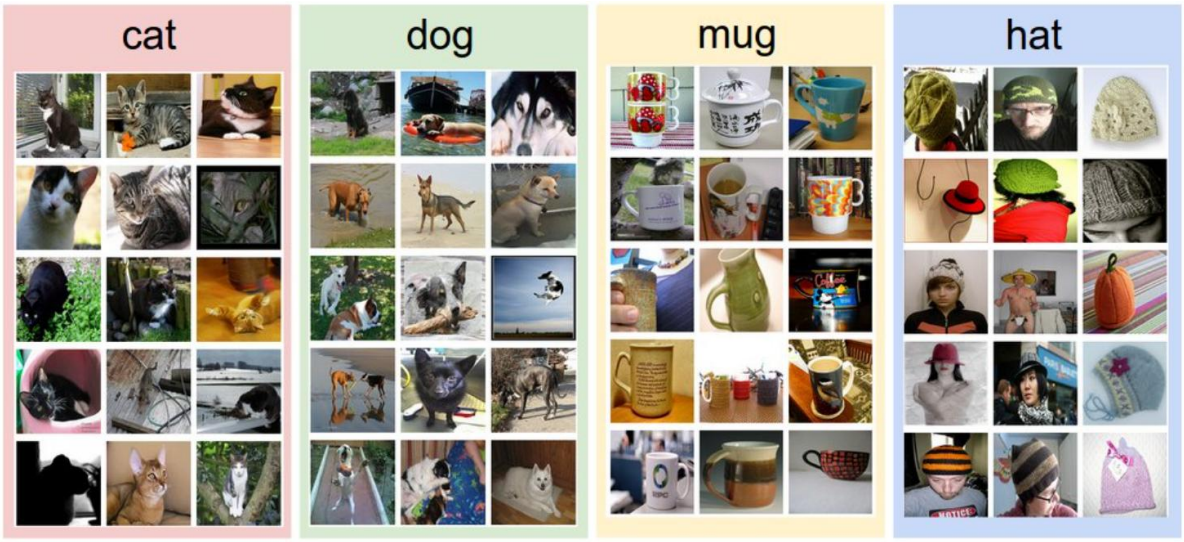


https://cs231n.github.io/classification/

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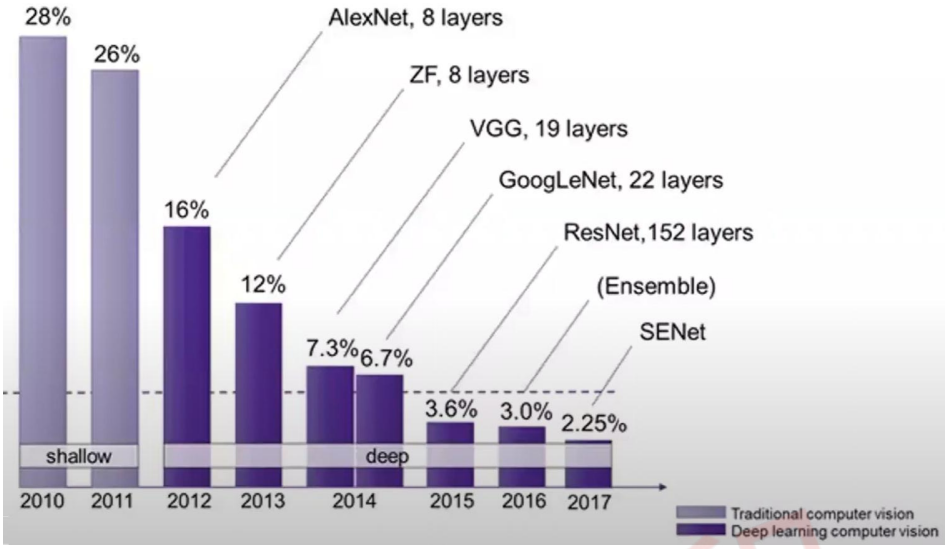
Data-driven approach



https://cs231n.github.io/classification/

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Datasets drive progress in computer vision



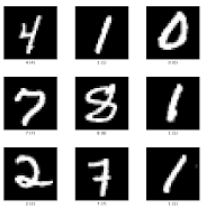
Courtesy David Held

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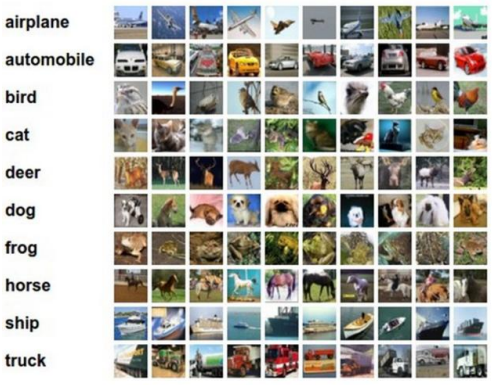
试用水印

Datasets

MNIST



CIFAR-10



ImageNet



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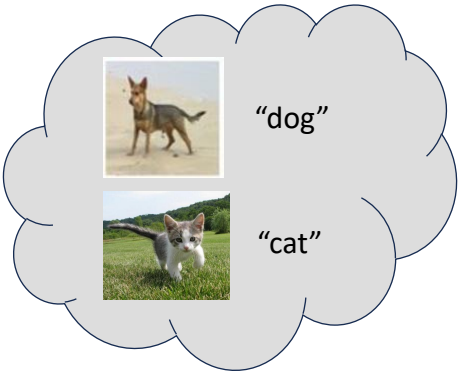
supervised - unsupervised - reinforcement learning

Supervised learning

Start with labeled dataset

Two steps:

1. **Train** classifier
2. **Inference** using trained classifier

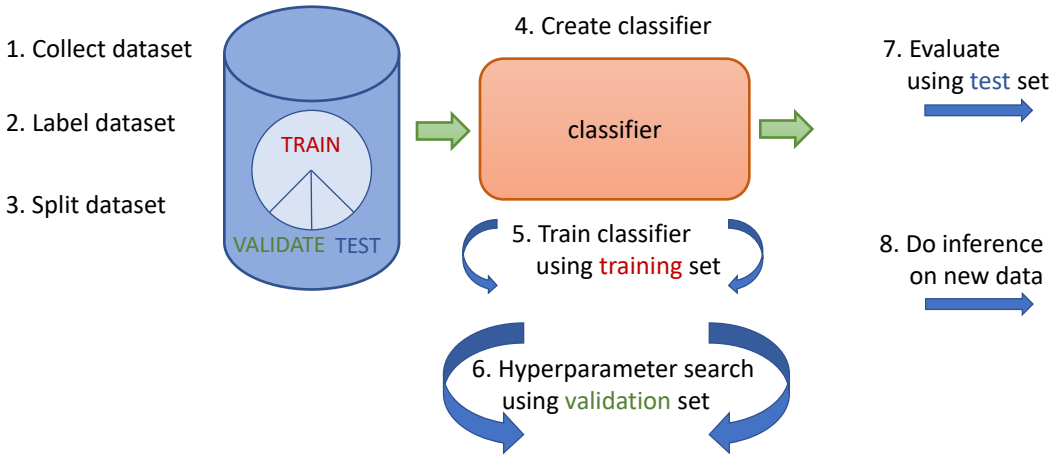


More detail in the next slide …

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Supervised learning



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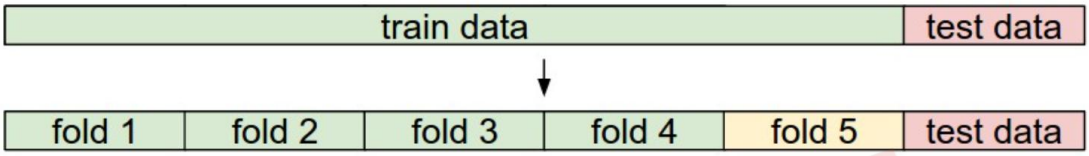
VALIDATE: find hyperparameters

Cross-validation

Split labeled data into
- training set
- validation set
- test set

Alternative: cross-validation

Rarely used these days b/c more computationally expensive



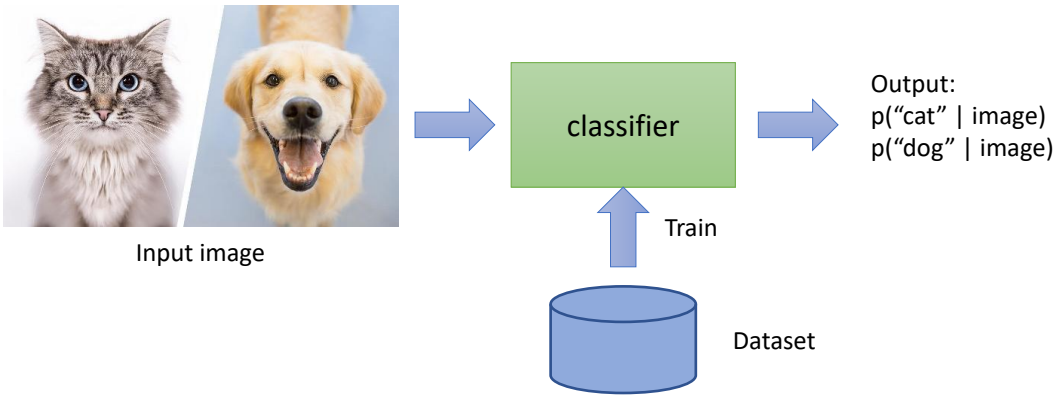
https://cs231n.github.io/classification/

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试用水印

Binary classification

Two categories



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Generative methods

ChatGPT

show an image of a cat



This is modeling the **likelihood**

p(image | “cat”)

Now, compare the input image with the set of all cat images

If distance is small, then it’s probably a cat

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试用水印

Nearest neighbors

This is the simplest generative method

Easy to understand, explain

Train: Simply store all the data

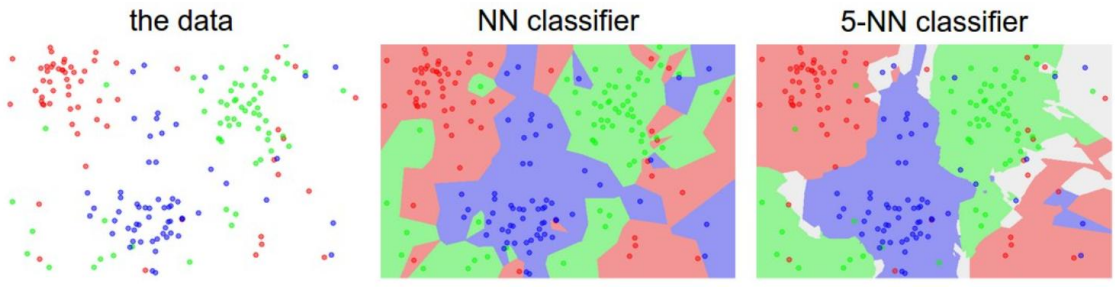
Inference: Find k nearest neighbors, do majority vote (k-NN)

Guaranteed to get better with more data

But

Does not scale well – computation and memory increase linearly with data

Requires distance metric



https://cs231n.github.io/classification/

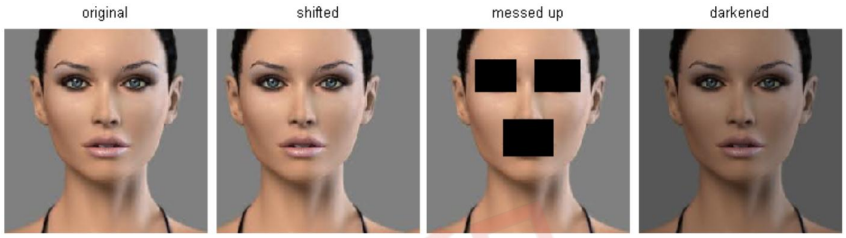
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L1 and L2 distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

test image					training image					pixel-wise absolute value differences				
56	32	10	18	-	10	20	24	17	=	46	12	14	1	→ 456
90	23	128	133		8	10	89	100		82	13	39	33	
24	26	178	200		12	16	178	170		12	10	0	30	
2	0	255	220		4	32	233	112		2	32	22	108	

$$d_2(I_1, I_2) = \sqrt{\sum_p (I_1^p - I_2^p)^2}$$

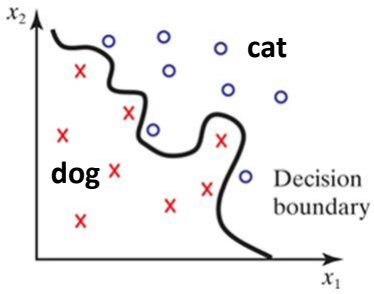


All these images are the same L_2 distance

https://cs231n.github.io/classification/

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Discriminative methods



Models the **posterior**

p("cat" | image)

All we care about is distance to the **decision boundary**

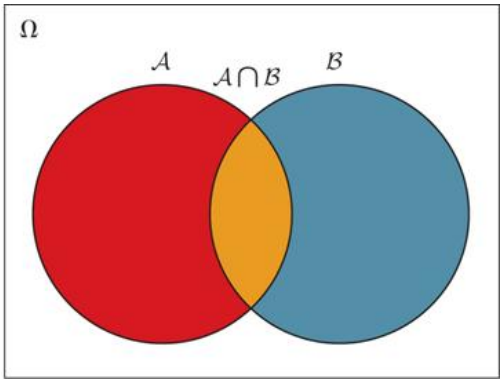
Shape of distribution doesn't matter

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Bayesian decision theory

How do **likelihood** and **posterior** relate to each other?

Bayes' rule



$$p(\mathcal{A}|\mathcal{B}) = \frac{p(\mathcal{B}|\mathcal{A})p(\mathcal{A})}{p(\mathcal{B})}$$

$$\text{posterior} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

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试用水印

Bayesian decision theory

How do **likelihood** and **posterior** relate to each other?

p("cat" | image)
posterior

p(image | "cat")
likelihood
(class-conditional density)

Bayes' rule

$$p(\omega_i|\mathbf{x}) = \frac{p(\mathbf{x}|\omega_i)p(\omega_i)}{p(\mathbf{x})}$$

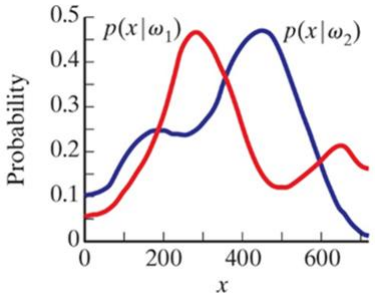
x is the feature
ω_i is the category

Don't be confused:
- Technically these are opposites
- But they perform similarly (because prior and evidence are *much* less important)

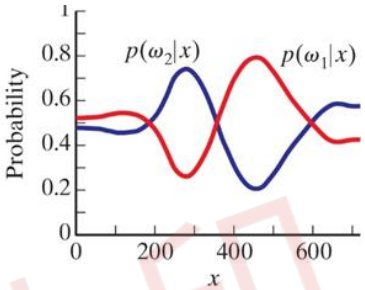
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Bayesian decision theory

Each **class-conditional density** integrates to 1



For each *x*, all **posterior** densities sum to 1



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Binary classification example



Training image



Ground truth labels

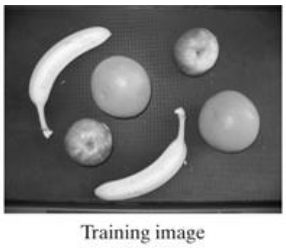
Label each pixel as foreground or background

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试用水印

Generative approach

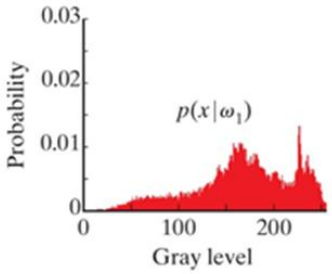
Compute histograms (non-parametric)



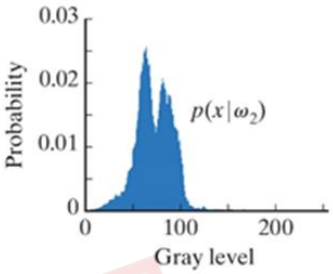
Training image



Ground truth labels



Foreground ω_1

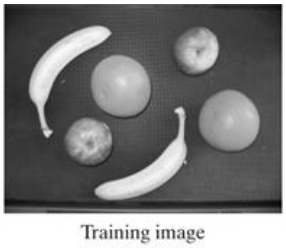


Background ω_2

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Generative approach

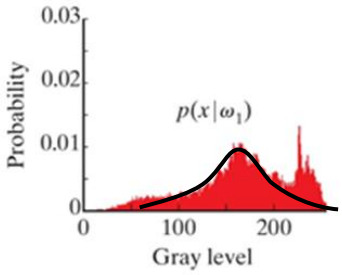
Can fit parametric distribution (Gaussian) if you want



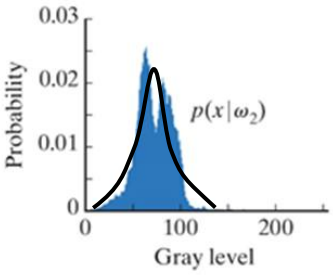
Training image



Ground truth labels



Foreground ω_1



Background ω_2

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试用水印

Discriminative approach



Training image

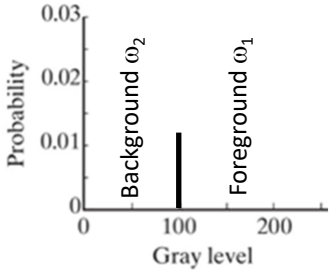


Ground truth labels

Distribution doesn't matter

If < **threshold**: background

If > **threshold**: foreground



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试用水印

Measuring errors

Confusion matrix:

	test positive (disease detected)	test negative (not detected)
actual positive (sick person)	9 (*TP*)	1 (*FN*)
actual negative (healthy person)	18 (*FP*)	72 (*TN*)

TPR, recall

FPR

precision

For rare events, *TN* is hard to measure (face detection)

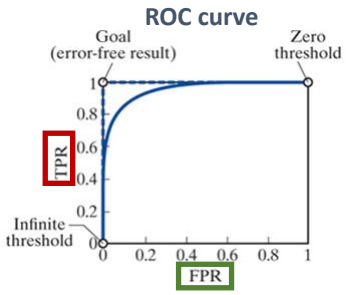
Metrics:

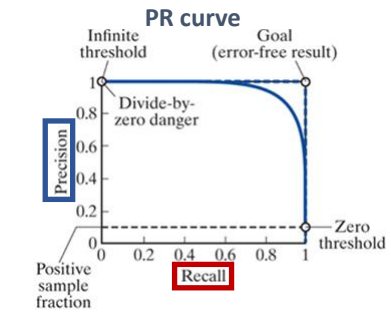
Name	Formula	Alternate names
true positive rate (*TPR*)	$\frac{TP}{TP + FN}$	recall, sensitivity, hit rate
false negative rate (*FNR*)	$\frac{FN}{TP + FN}$	miss rate
true negative rate (*TNR*)	$\frac{TN}{FP + TN}$	specificity
false positive rate (*FPR*)	$\frac{FP}{FP + TN}$	false alarm rate, fallout
precision	$\frac{TP}{TP + FP}$	positive predictive value
accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	

$TPR + FNR = 1$

$TNR + FPR = 1$

PR curve is better for imbalanced dataset





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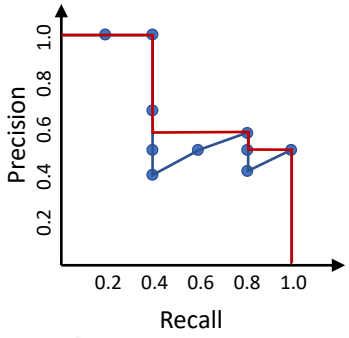
Example: PR curve

Suppose we have ten items to be classified

Unordered:		Reordered:	
−	0.71	+	0.95
+	0.34	+	0.90
−	0.76	−	0.87
+	0.68	−	0.76
−	0.87	−	0.71
+	0.95	+	0.68
+	0.90	+	0.53
−	0.39	−	0.47
−	0.47	−	0.39
+	0.53	+	0.34
Data		Output	

Rank	Correct?	TP	FP	FN	Precision ($\frac{TP}{TP+FP}$)	Recall ($\frac{TP}{TP+FN}$)
1	True	1	0	4	1.0	0.2
2	True	2	0	3	1.0	0.4
3	False	2	1	3	0.67	0.4
4	False	2	2	3	0.5	0.4
5	False	2	3	3	0.4	0.4
6	True	3	3	2	0.5	0.6
7	True	4	3	1	0.57	0.8
8	False	4	4	1	0.5	0.8
9	False	4	5	1	0.44	0.8
10	True	5	5	0	0.5	1.0

5 are apples +
5 are not apples −



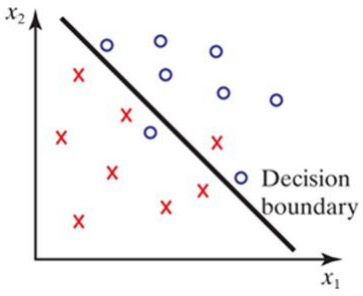
It is common to smooth the curve

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Bias-variance tradeoff

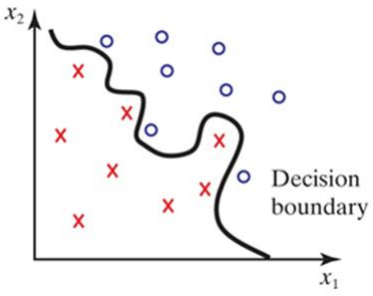
Bias – How poorly model fits training data
Variance – How sensitive the model is to the particular choice of training dataset

Underfit



High bias
Low variance

Overfit



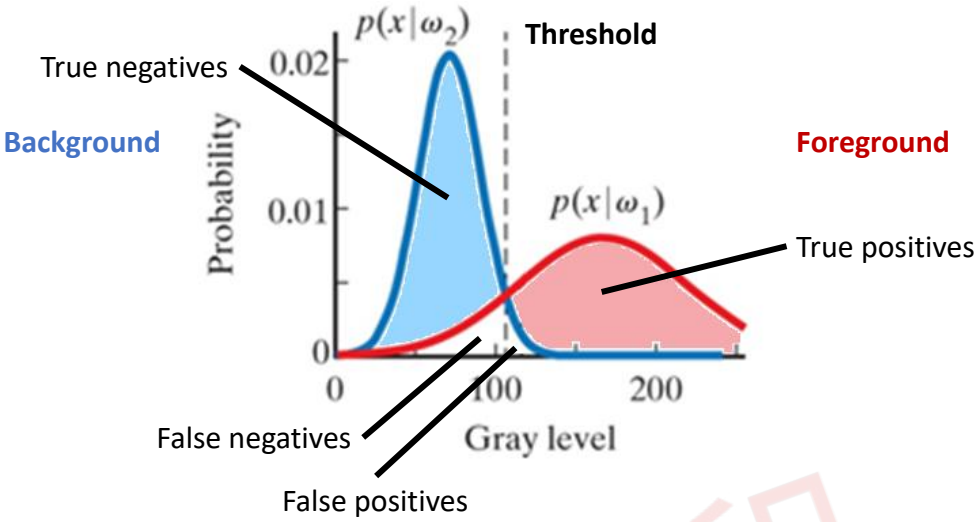
Low bias
High variance

Occam's razor: "The simplest explanation is the best"

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试用水印

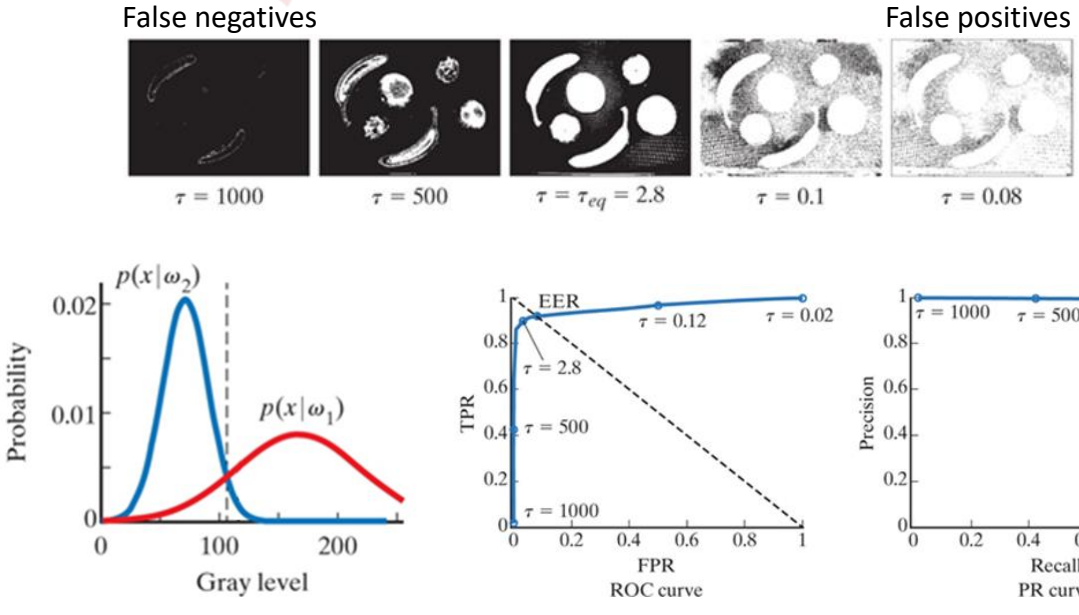
Back to our example…



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试用水印

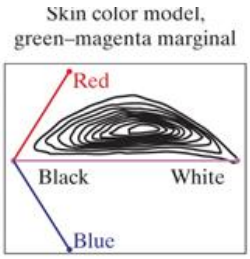
Back to our example…

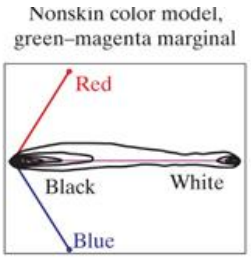


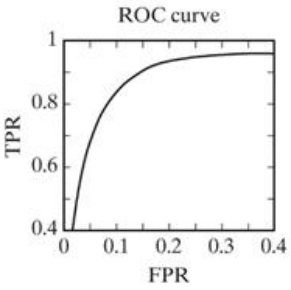
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Multidimensional features

Generative (histogram)





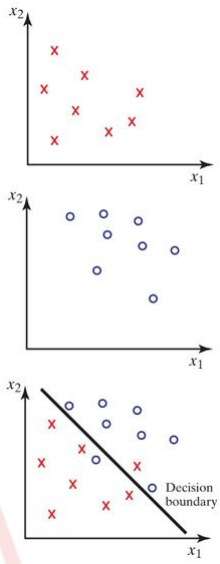








Discriminative

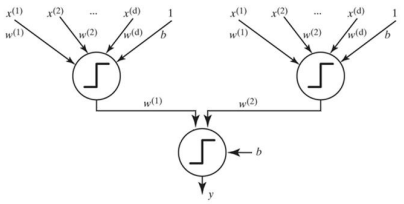


Jones and Rehg, 1998, https://www.hpl.hp.com/techreports/Compaq-DEC/CRL-98-11.pdf

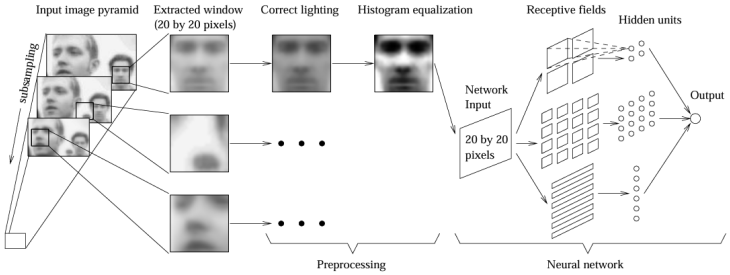
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Neural network

Simple idea, scales well with data



Rowley-Baluja-Kanade 1996
Face detection

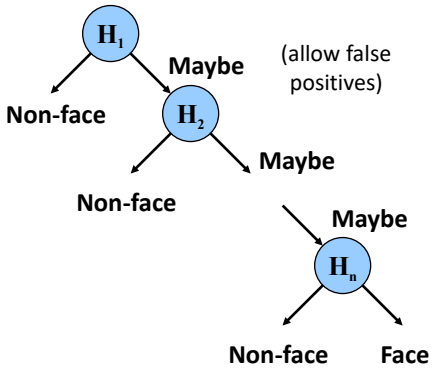


Deep learning is based on this

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Decision tree

Can be extremely efficient



(allow false positives)



Viola-Jones 2001

Cascade classifier

https://cs231n.github.io/classification/

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试用水印